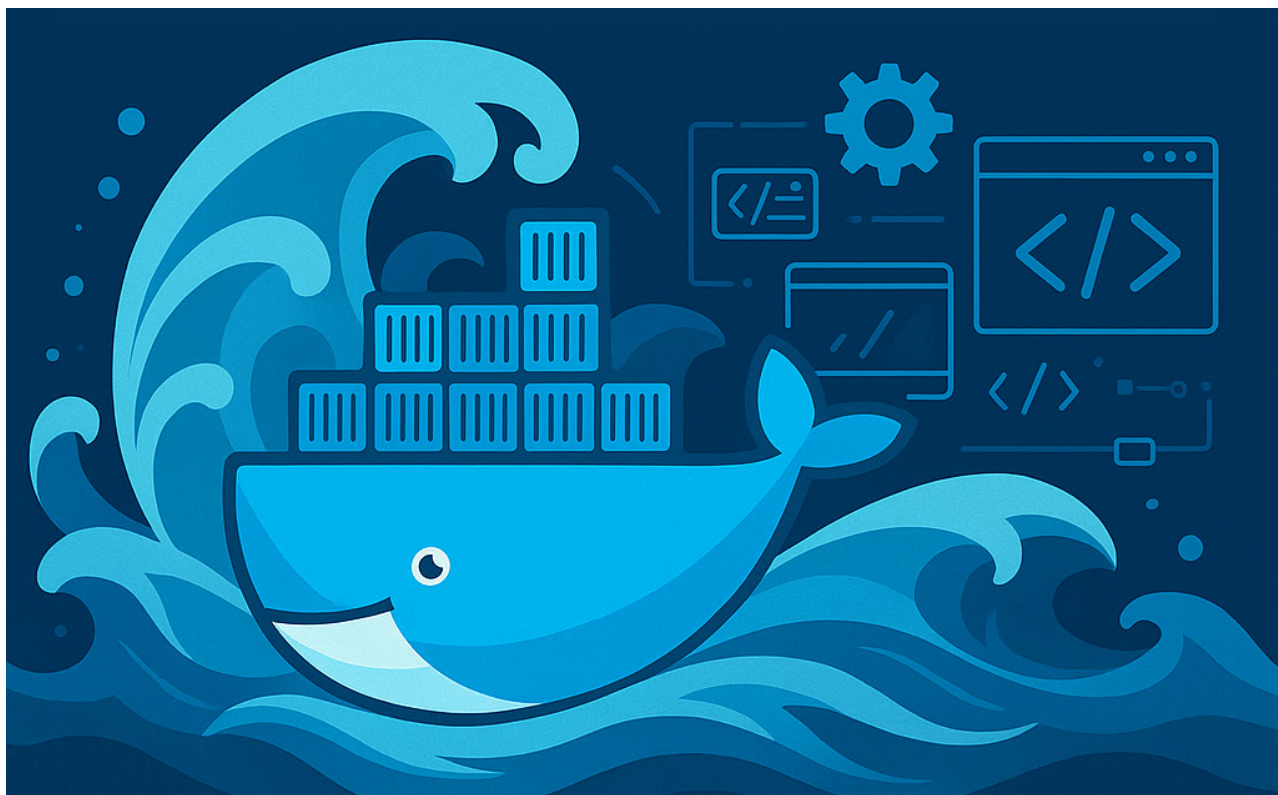


Docker: Powering the Next Wave of Software Development

In a world where organisations are transforming their infrastructure to house AI-based solutions, Docker and Kubernetes are proving to be powerhouses for developing secure and scalable software that is delivered with speed.



In October 2023, Docker announced the release of Docker AI — a revolutionary product within its developer toolkit that brought smart help directly into the software development process. Built to drive AI/ML application productivity and security to its optimal level, Docker AI is much more than code generation. Whereas tools such as GitHub Copilot and Tabnine aid in writing source code — which is generally only 10–15% of an application — Docker AI is taking on the larger and more nuanced 85–90% of app development, which is made up of infrastructure elements such as Dockerfiles, language runtimes, databases, message queues, and configuration files. This is important because real-world applications aren't solely about logic and algorithms but about the entire system they inhabit.

Docker AI offers developers context-aware, automated advice while they're coding Dockerfiles, creating Compose files, or debugging builds. It taps into the experience of millions of developers across the Docker ecosystem and leverages collectively shared best practices from Docker Hub, GitHub, and other community repositories. With Docker CEO Scott Johnston noting that Docker AI speeds up the whole development “inner loop,” developers can fix issues and iterate faster with more confidence. Backed by community-tested findings from the broader Docker community, the AI provides recommendations that are not only optimised but also secure and production-ready. According to Katie Norton of IDC, this type of generative tooling enhances productivity and developer happiness while reducing operational workloads.